

SKY CLOUD IDENTIFICATION PROJECT

EXTRACT PATTERNS FROM CLOUD IMAGES & PREDICT SHORT-TERM WEATHER TRENDS

ATMOSPHERIC DATA



TEMP 22°C
HUMID 65%
WIND 12 km/h
VIS 30 km

CIRRIFORMS

High altitude, thin and wispy,
composed of ice crystals.



CUMULIFORMS

Puffy, vertical development,
formed by rising warm air.



STRATOCUMULIFORMS

Lumpy, patchy layers,
intermediate altitude.



STRATIFORMS

Uniform layers, low altitude,
often covering large areas.



DATA NODE

SKY OBSERVATION SYSTEM

ACTIVATED BY
MARTHA BERKHOLZ, AMAL RAMIA & CONGYI HU



Why Cloud Type identification???



Weather
Forecasting

Climate
Research

Aviation

Solar
Energy

Automation



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Project Goal

Input

Labeled sky photos

- Images
- cloud mask annotations

**Train a Mask
R-CNN model to
learn:**

- where clouds are
- which cloud type each region is

Output

- cloud locations
- cloud class labels
- cloud segmentation masks



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Literature Review

Study	Objective	Model	Dataset Size	Main Result
Deep Learning-Based All-Sky Cloud Image Recognition	Cloud type recognition	CNN	256 images	95–100% accuracy
Artificial Neural Networks in Automatic Image Classification of Clouds	Cloud recognition from ground photographs	CNN	200,000 images	97.4% accuracy
CNN for High-Resolution Cloud Motion Prediction	Predict cloud movement	CNN	150,000 images	94% (1 min prediction)
Cloud Identification from All-Sky Camera Data with Machine Learning	Detect opaque clouds	ResNet, LightGBM	2,000 / 1,000 images	85% (ResNet), 95% (LightGBM)
Deep Learning-Based Identification of Precipitation Clouds	Detect rain-related clouds	EfficientNet, LeNet, DeiT, AlexNet	2,445 images	99% rainfall detection
Cloud Detection in Sky Images (GitHub)	Detect cloud pixels	NRBR + CSL algorithm	Not specified	Traditional approach, DL recommended

Dataset Overview

Dataset

- Extension of the **Clouds-1000** dataset
- Source: Universidade Federal de Santa Catarina

Images

- 996 annotated sky images
- Resolution: **2592 × 1944 px**
- Collected between **March – October 2021**

Task

- Multi-class cloud identification
- **Polygon annotations** for each cloud region



Example Picture from the Dataset



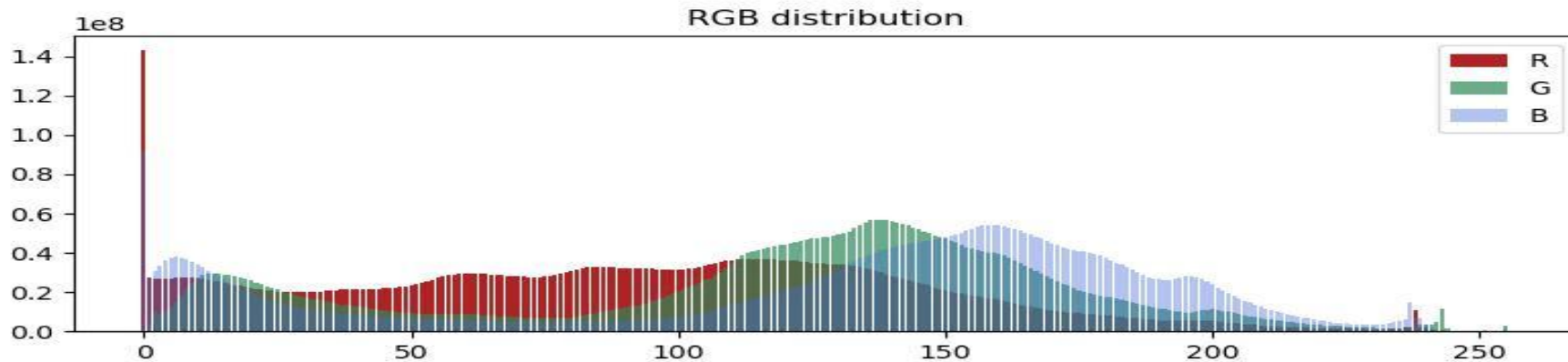
Feature Description

Cloud Type	Amount in Dataset	% in Dataset
Arvore	989	99.30%
Estratocumuliformes	812	81.53%
Estratiformes	271	27.21%
Cirriiformes	285	28.61%
Cumuliformes	90	9.04%

- ❑ **Feature 1 (Arvore):** representation of other objects (trees, buildings)
- ❑ **Feature 2 (Stratocumuliformes):** lower clouds (1-4km), very humid
- ❑ **Feature 3 (Stratiformes):** horizontal, appear in stratified atmosphere
- ❑ **Feature 4 (Cirriiformes):** high clouds
- ❑ **Feature 5 (Cumuliformes):** = Cumulonimbus, formed in dryer regions, therefore rarely presented in this dataset (mountainside in brazil)



Dataset Characteristics



RGB Characteristics

- High amount of blue pixels (difference to ImageNet Datasource)
- Low contrast

Implementation in Training

- > freeze only layer 1, fine-tune from layer 2
- > CLAHE augmentation



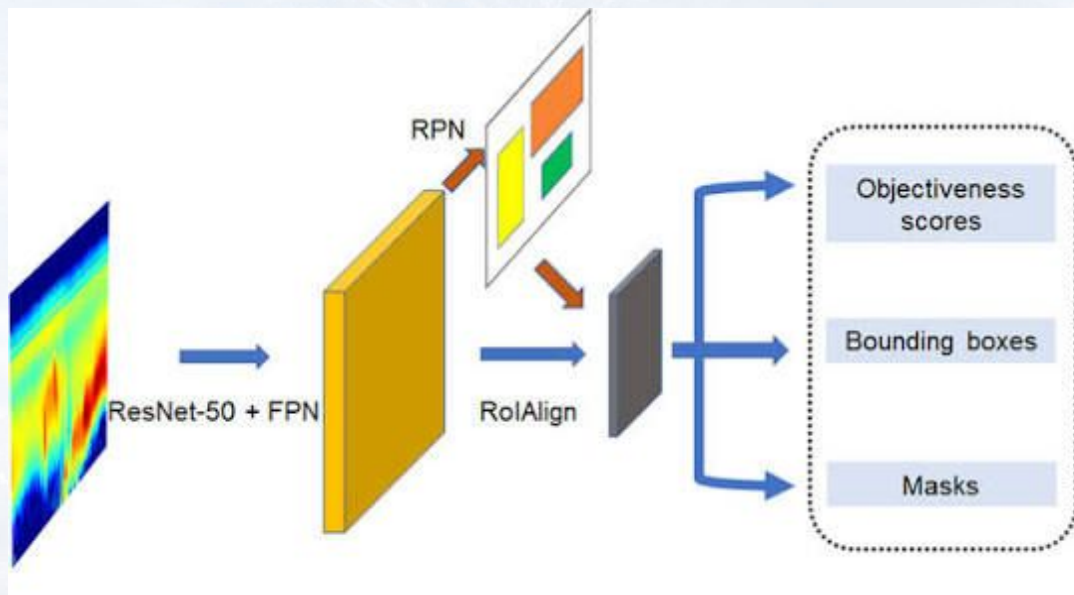
Baseline Model - Faster R-CNN

Amount of Training Data and Quality of Outcome

Samples	maP
50	0.1832
100	0.2214
300	0.2848
500	0.3340
796	0.3417



Main Model - Mask R-CNN ResNet-50 FPN



Model Type

- Instance Segmentation

Backbone

- ResNet-50 + Feature Pyramid Network (FPN)

Why this model?

- Uses polygon annotations
- Handles irregular cloud shapes
- Detects multiple cloud regions in one image



Training Setup

Framework	PyTorch + Torchvision (Google Colab, GPU/T4)
Input prep	Images loaded from <code>images_small</code> Polygon annotations (JSON) converted to masks + bounding boxes Coordinates scaled to 224×224 , 512×768
Data augmentation & normalization	<code>ColorJitter</code> + random horizontal flip ImageNet normalization Clahe Augmentation <code>torchvision.transforms.v2</code>
Dataset split (from notebook output):	Train: 747 Validation: 124 Test: 125
Optimization	SGD (<code>lr=0.01</code> , <code>momentum 0.9</code> , <code>weight decay 0.0005</code>) CosineAnnealingLR, up to 20 epochs Early stopping (<code>patience=5</code> , <code>min_delta=0.001</code>)

Results I

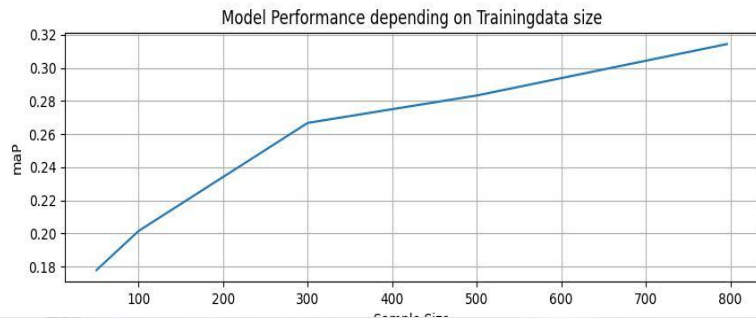
Baseline Model Fast RCNN



Mask RCNN only layer 1 frozen



Mask RCNN w/ Clahe, only layer 1 frozen

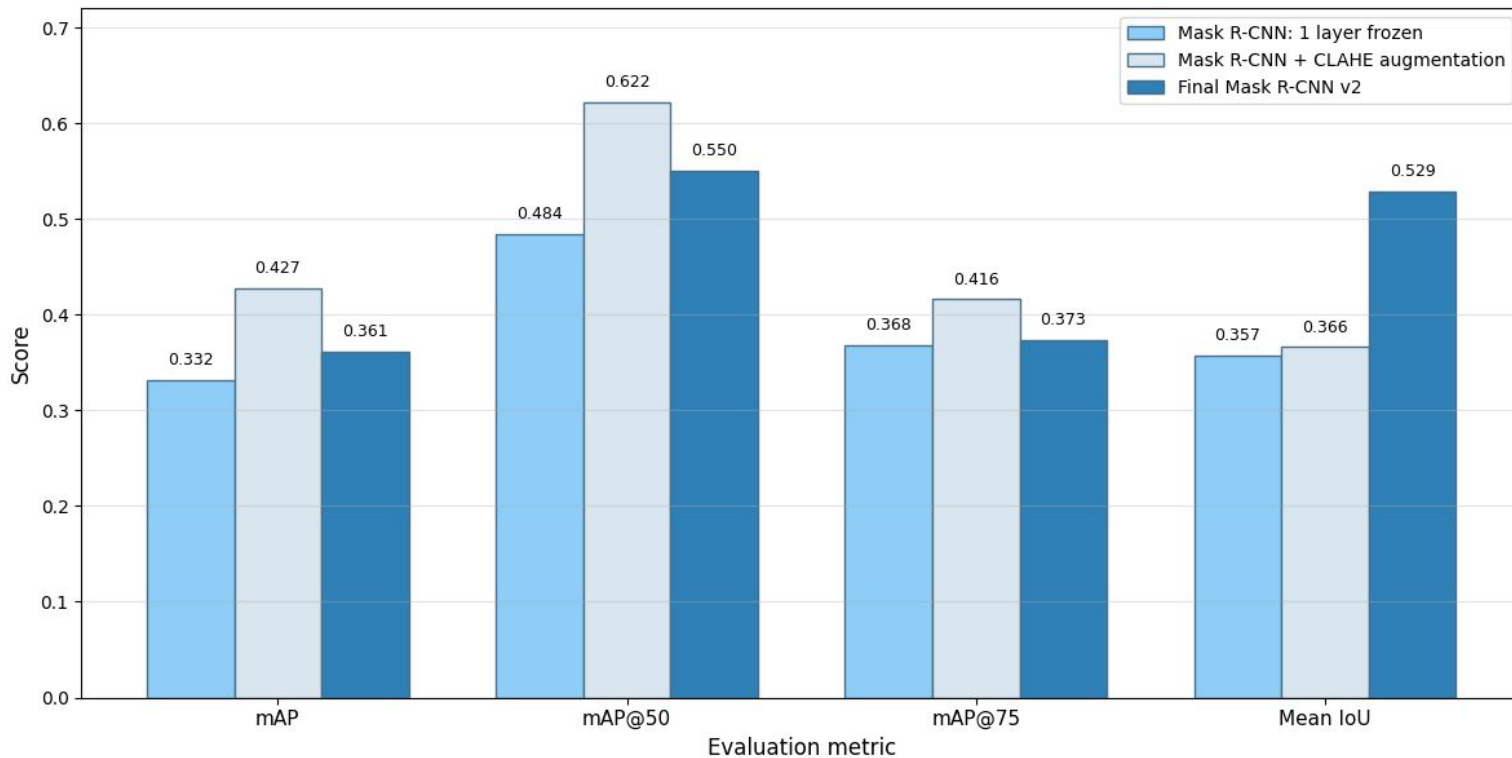


Mask RCNN v2 layer 1 and 2 frozen



Results II

Comparison of Mask R-CNN Model Performance



Conclusion and Future Work

Mask R-CNN with **CLAHE** achieved the best detection performance: highest mAP, mAP@50, and mAP@75.

Final Mask R-CNN V2 achieved the highest Mean IoU, meaning its predicted masks overlapped the ground-truth masks more accurately on average.

CLAHE improved cloud visibility and detection, while **Final Mask R-CNN V2** improved segmentation mask quality.

The **basic Mask R-CNN** model performed reasonably, but it was outperformed by the **CLAHE** model in detection metrics and by the **V2** model in mask overlap quality.

IOU & mAP misalignment due to:

- missed clouds
- wrong classes
- extra false predictions
- low-confidence correct predictions...

A pipeline combined with
→ **Mask R-CNN ResNet50 FPN V2** model
→ **CLAHE** preprocessing
→ **Torchvision.transforms.v2** augmentation/resizing

Challenges and Errors

→ Initial approach: Faster R-CNN

- ◆ Bounding boxes were not suitable for irregular cloud shapes and often included background objects.

→ Data preparation

- ◆ Images, boxes, and masks had to remain synchronized during resizing and augmentation.(e.g. problem with RandAugment)

→ During Training:

- ◆ Performance had to be balanced with GPU limits, frozen layers, dataset size, and number of epochs.

→ Optimization:

- ◆ Learning rate and augmentation settings strongly affected mAP and model stability.



Discussion

Google Lens
can do the
same ?



Small dataset
limits the
performance?



Dataset
universally
applicable ?



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Q & A

Thank you for listening!!

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